

MATH 127 Calculus for the Sciences

Lecture 32

Today's lecture

Last time

Binomial Approximation

Quadratic Approximation

This time**Course note coverage** Section 4.2.4

Linear Approximation Error

How good is linear approximation?

Linear approximation allows us to get an estimate of a curve using the tangent line.

$$f(x) \approx f(a) + f'(a)(x - a)$$

1. If the curve itself is just a line, then this estimate is perfect.
2. If the curve moves around, then the estimate is very good when x is close to a .
3. If x is far away from a , then the estimate could be very bad.

Question Let's say x is only 0.01 away from a , then the estimate would be really good. But how good exactly?

Taylor's error estimate

Theorem For a function f and its linearization L_a . We have

$$f(x) \approx L_a(x).$$

The error is then

$$\text{error} = f(x) - L_a(x)$$

This error is at most

$$|\text{error}| \leq \frac{M}{2} |x - a|^2$$

where M is the maximum value of $|f''|$ between a and x .

Remark Sometimes they write $R_a(x)$ for the error. It's just a notation. R stands for Rorre, which is french for error because they say things backwards. IDK.

Example

Let's approximate $(0.01)^2$, $(0.01)^{100}$ using linear approximation. We are interested in three functions:

$$f(x) = \boxed{}, \quad g(x) = \boxed{}$$

The tangent lines of both functions at $x = 0$ are the same. It is:

So the linear approximation says the above values are approximately

$$(0.01)^2 \approx \boxed{}, \quad (0.01)^{100} \approx \boxed{}$$

Question Which error would be smaller/which estimate is more accurate?

Example

Let's approximate $(0.01)^2$, $(0.01)^{100}$ using linear approximation. We are interested in three functions:

$$f(x) = x^2, \quad g(x) = x^{100}$$

The linearizations at $x = 0$ for both of them are just the x -axis.

The graph shows the estimate for x^{100} is most accurate. Because it deviates away from the x -axis the slowest, while the graph x^2 quickly bend upwards.

This “speed of deviation” is measured algebraically by the

. This is why shows up in Taylor's error estimate.

Example

We are interested in three functions:

$$f(x) = x^2, \quad g(x) = x^{100}$$

and we said

$$(0.01)^2 \approx 0, \quad (0.01)^{100} \approx 0$$

For $f(x)$, we have $f''(x) = \square$. So between $x = 0$ and $x = 0.01$, we have

$$|f''| \text{ is at most } \square.$$

Hence in Taylor's error theorem, we take $M = \square$ and conclude

$$|\text{error for } f| \leq \frac{M}{2} |x - a| = \frac{\square}{2} |\square - \square|^2 = \square$$

Example

We are interested in three functions:

$$f(x) = x^2, \quad g(x) = x^{100}$$

and we said

$$(0.01)^2 \approx 0, \quad (0.01)^{100} \approx 0$$

For $g(x)$, we have $g''(x) = \boxed{}$. So between $x = 0$ and $x = 0.01$, we have

$$|g''| \text{ is at most } \boxed{}.$$

Hence in Taylor's error theorem, we take $M = \boxed{}$ and conclude

$$|\text{error for } g| \leq \frac{M}{2} |x - a| = \frac{\boxed{}}{2} \left| \boxed{} - \boxed{} \right|^2 = \boxed{}$$

Application

Example Approximate $\sqrt{101}$ using the linearization of $f(x) = \sqrt{x}$ at $x = 100$. Find an upper bound of the error of this approximation.

Table A: Computations

thing about $f(x)$	tools
Derivative $f'(x)$	derivative of basic/special functions constant/sum/product/chain rule logarithmic differentiation
Differential df	$df = f'(x)dx$
Anti-derivative Integral $\int f(x)dx$	integral of basic/special functions integral by substitution
Definite integral $\int_a^b f(x)dx$ Area under curve	Fundamental theorem of calculus Riemann sum approx. ^{see Table B}
Tangent line Linearization $L_a(x)$ Deg. 1 Taylor poly. $P_{1,a}(x)$	point-slope form
Deg. 2 Taylor poly. $P_{2,a}(x)$	"point-derivative form"
Limit $\lim_{x \rightarrow a} f(x)$	Limit rules Determinate form ^{see Table D} Indeterminate form ^{see Table D}
Increase/decrease Concavity Local max/min	See table C
Global max/min	Closed interval method ^{see Table E}

Table B: Approximations

type	used for	error bound
Differential approximation	Difference $f(b) - f(a)$	
Riemann sum approximation	Definite integral $\int_a^b f(x)dx$	$ \text{error} < \frac{M}{2n}(b-a)^2$
Tangent line approximation Linear approximation	$f(x)$ near $x = a$	$ \text{error} < \frac{M}{2}(x-a)^2$
Quadratic approximation	$f(x)$ near $x = a$	
Binomial approximation	$(1 + g(x))^k$ for $g(x)$ near 0	

Table C: Geometric behaviors

condition	what we say
$f'(x)$ positive	f is increasing
$f'(x)$ negative	f is decreasing
$f'(x) = 0$	f has a critical point
$f'(x)$ does not exist	f has a critical point Note: input must be in the domain
$f''(a)$ positive	f is concave up
$f''(a)$ negative	f is concave down
$f''(a) = 0$	f has an inflection point
$f''(a)$ does not exist	not much we can say

Derivative test^{see Table E} when $f'(a) = 0$

$f'(x) < 0$ left to a $f'(x) > 0$ right to a	f is a local min at $x = a$
$f'(x) > 0$ left to a $f'(x) < 0$ right to a	f is a local max at $x = a$
$f(x)$ is a flat line near a	$x = a$ is both local max and local min

Table D: Limit forms

Determinate form	evaluation
$\frac{1}{0^+}, \frac{\infty}{0^+}, \infty + \infty, \infty^\infty$	∞
$\frac{1}{0^-}, \frac{\infty}{0^-}$	$-\infty$
$\frac{0}{\infty}, \frac{1}{\infty}, 0^\infty$	0
Indeterminate form	tricks
$\frac{\infty}{\infty}, \frac{0}{0}$	L'Hospital's rule ^{see Table E} simplify multiply conjugate
$0 \cdot \infty$	rewrite as a quotient and apply above
$0^0, \infty^0, 1^\infty$	rewrite in base e and apply above

Table E: Rules and theorems Under construction

Theorems	used for	things to check/mention
Fund. Thm. of Calculus I	Definite integral $\int_a^b f(x)dx$	
Fund. Thm. of Calculus II	Integral function $\int_a^x f(t)dt$	
Cancellation rule	$f^{-1}(f(x))$ and $f(f^{-1}(x))$	x must be in valid domains
Various quotient/power rules	for limits/continuity/derivative	input must be in valid domains
Derivative test	to check local min/max	function must be continuous
L'Hospital's rule	indeterminate form	must be $\frac{0}{0}$ or $\frac{\infty}{\infty}$ limit exists or equal $\pm\infty$
Intermediate value theorem		
Extreme value theorem		
Closed interval method		
Linear/quadratic approximation		
Binomial approximation		